Iron metabolism

Chemical forms of iron :-

- Ferric (Fe +3) ⇝ insoluble in Physiological PH
- Ferrous (Fe +2) ⇝ daengorus if free , forms free radical

Iron in Heme Proteins :-

1. Chytochrome ⇝ mytochondria
2. Heamoglobin
3. myoglobin ⇝ muscle oxgen storage Protein
4. Chytochrome P-450 ⇝ liver ⇒ deloxify foreign substances

non- Heme iron Proteins :-

1. Ferritin ⇝ iron storage Protein
2. transferrin ⇝ iron transport Protein

amount of body iron stores :-

- men ⇒ (20 - 275 Mg/L)
- women ⇒ (5-200 Mg/L)

15 Mg/L and less → insuficient iron store.

Disorders of Iron metabolism :-

- Haemochromatosis ⇝ increased absorption of iron from the gut
- Iron deficiency anemia ⇝ decresed amount of iron in the body
- anemia of chronic disease ⇝ inflamation - induced change of iron distribution

Couses of iron deficiency :-

1. Chronic blood loss
2. Incresed demand
3. malabsorbtion of iron
4. inadequate iron intake
5. Vascular hemolysis
6. Combination

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Lab Features :

Hb, Htc, Rbc : low

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MCV, MCH, MCHC : low

RDW : High

Retics : normal / low

Plt : Normal / low / high

WBC : normal / low

Serum iron : low

TIBC : High

Serum ferritin : low

transferrin saturation : low

Serum transferrin receptor : high

free erythrocyte protoporphyrin : high

Ferroportin : transmembrane protein that export iron from the cells into blood stream.

which cells must be able to export iron ?

- macrophages
- enterocytes
- Hepatocytes

Hepcidin : blocks iron export from macrophages and entrocytes

* released from the liver according to body iron status :

- iron overload increses hepcidin
- iron deficiency decreases hepcidin expression
- inflamation increse hepcidin production